

Basic Electrical Engineering (BT EE103A)

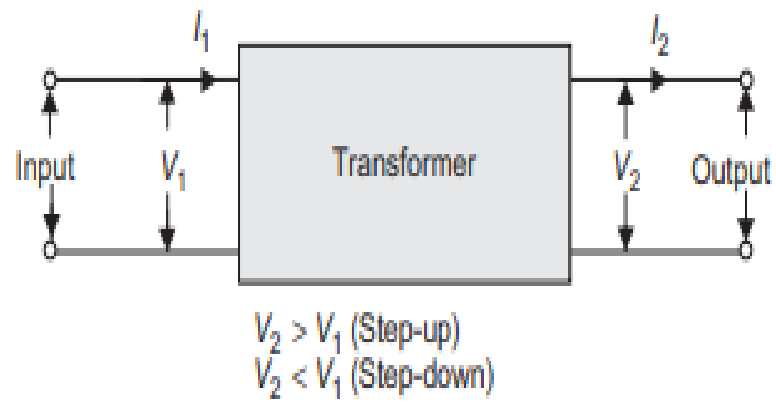


हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

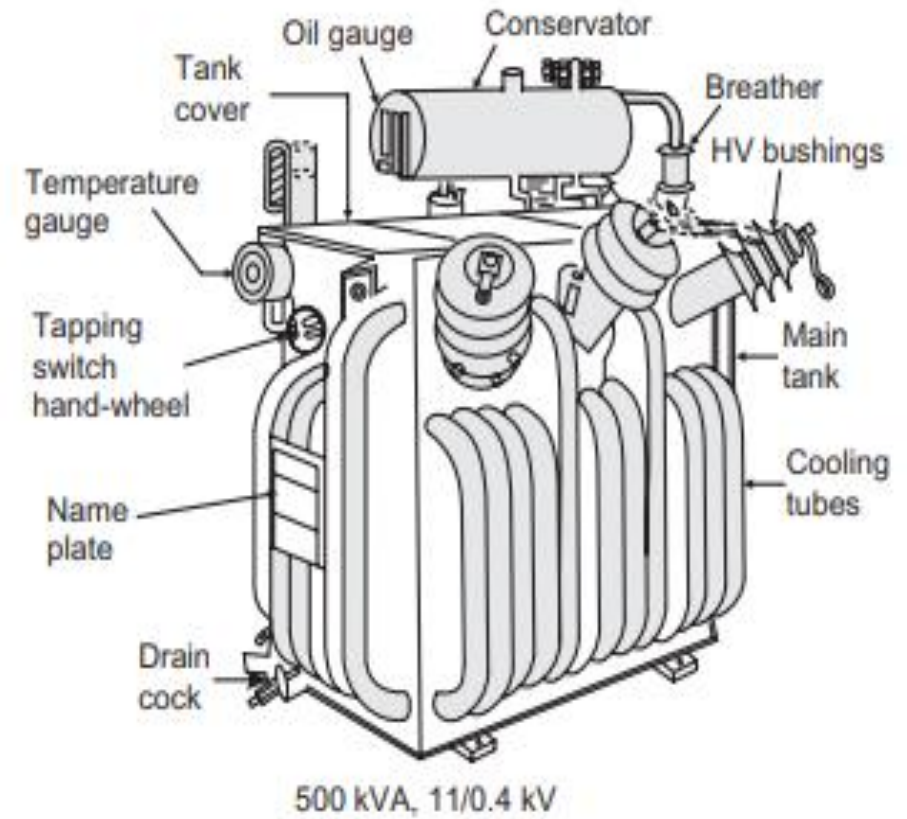
Unit-III

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TRANSFORMER

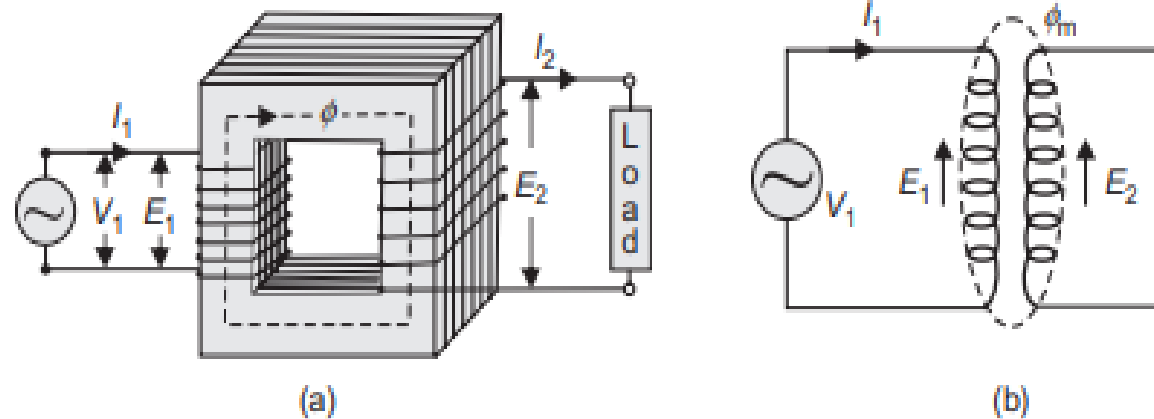


Block diagram of a transformer



WORKING PRINCIPLE OF A TRANSFORMER

The basic principle of a transformer is electromagnetic induction



(a) Simple diagram of a transformer (b) Mutual flux linking with primary and secondary winding

The induced emf in the primary and secondary winding depends upon the rate of change of flux linkages, that is, $\left(N \frac{d\phi}{dt}\right)$. The rate of change of flux ($d\phi/dt$) is the same for both primary and secondary windings. Therefore, an induced emf in primary winding is proportional to number of turns of the primary winding ($E_1 \propto N_1$), and in secondary winding, it is proportional to number of turns of the secondary winding ($E_2 \propto N_2$).

$\therefore$ In case $N_2 > N_1$, the transformer is step-up transformer, and when $N_2 < N_1$, the transformer is step-down transformer.

Turn ratio: The ratio of primary to secondary turns is called turn ratio, that is, turn ratio = N_2/N_1 .

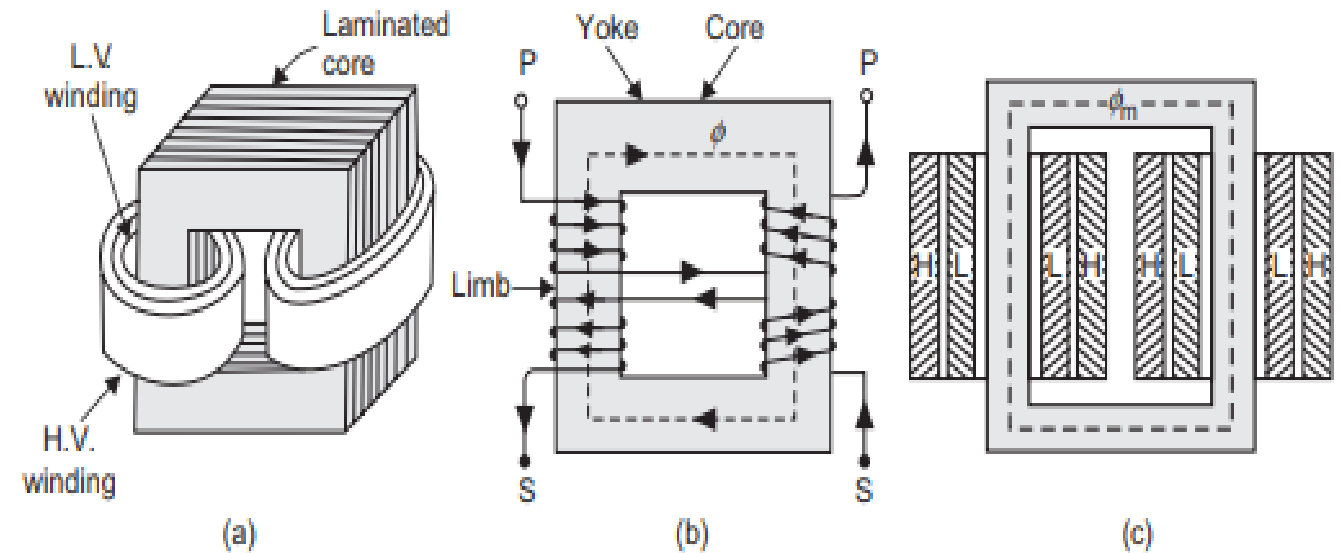
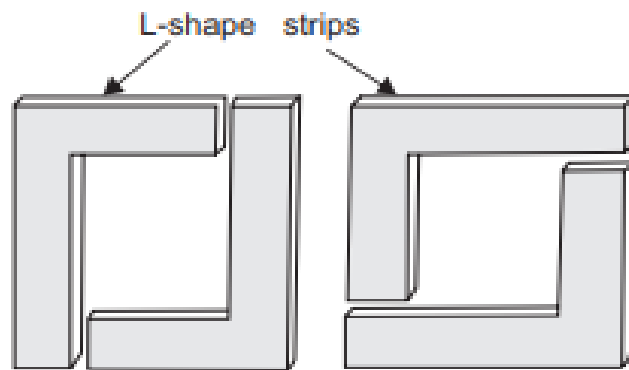
Transformation ratio: The ratio of secondary voltage to primary voltage is called voltage transformation ratio of the transformer. It is represented by K .

$$K = \frac{E_2}{E_1} = \frac{N_2}{N_1} \text{ (since } E_2 \propto N_2 \text{ and } E_1 \propto N_1 \text{)}$$

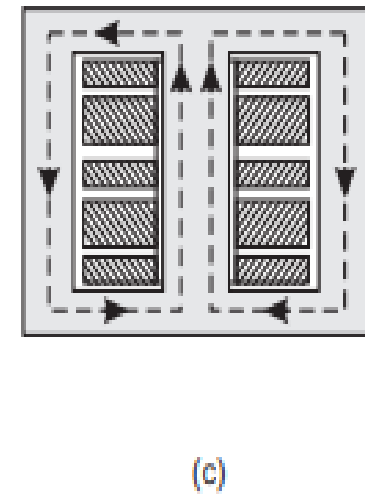
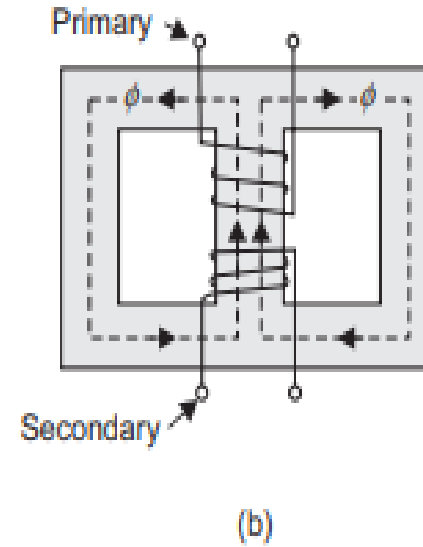
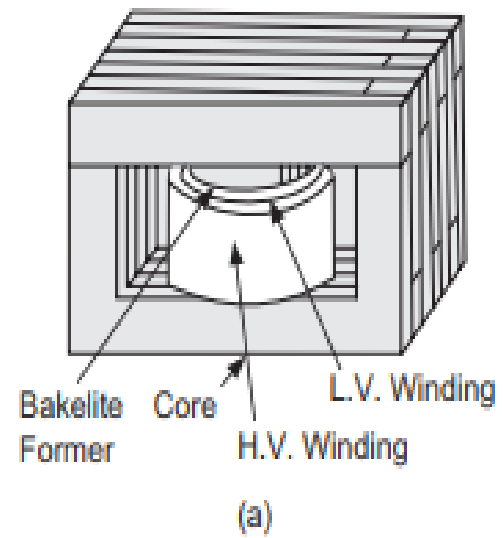
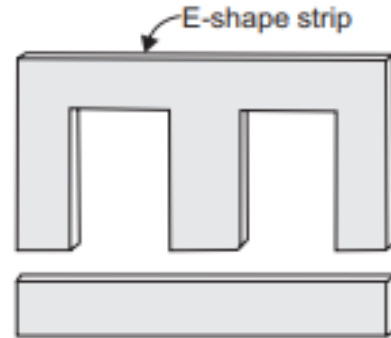
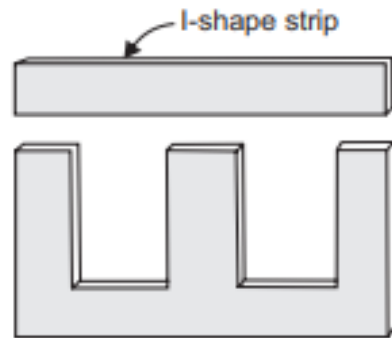
CONSTRUCTION OF A SINGLE-PHASE SMALL RATING TRANSFORMER

1. Core-type transformers
2. Shell-type transformers
3. Berry-type transformers

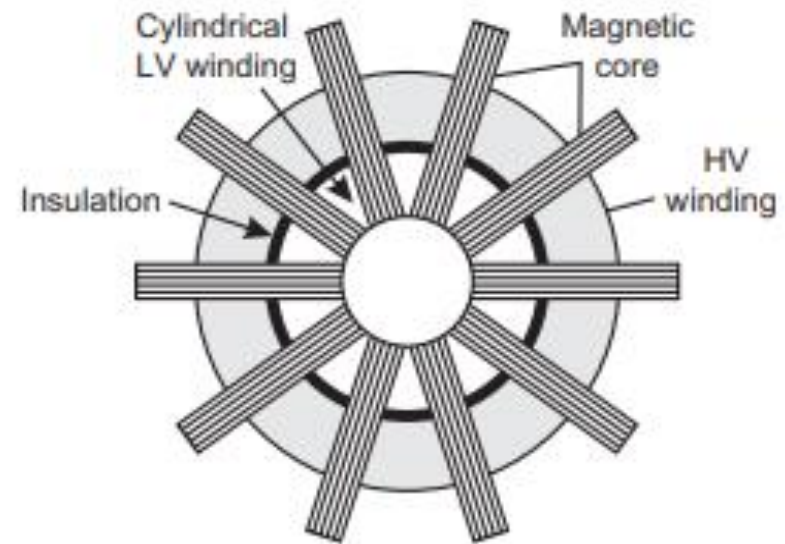
1. Core-type transformers



2. Shell-type Transformers

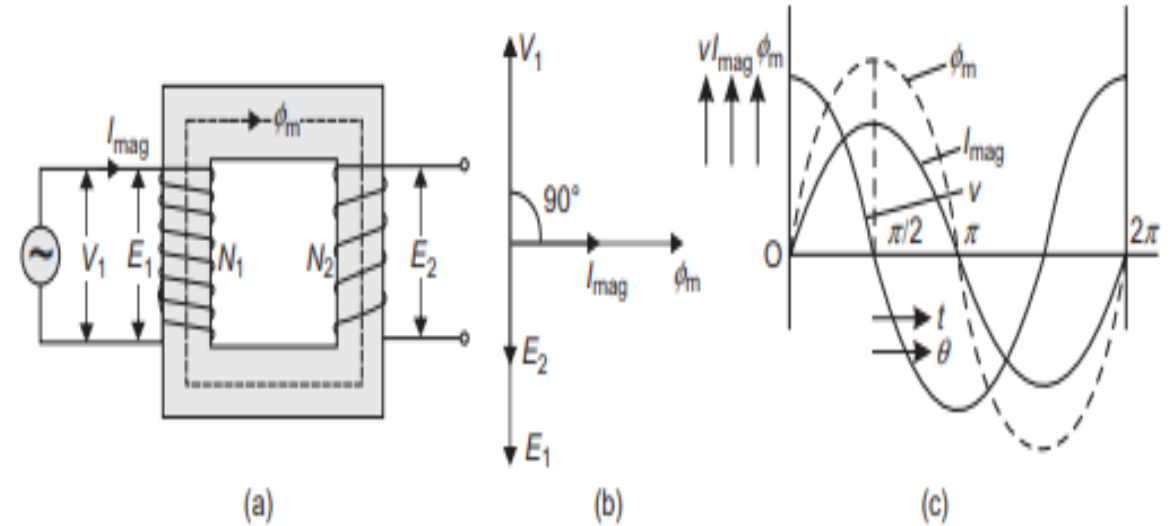


Berry-type Transformers



AN IDEAL TRANSFORMER

1. Its coefficient of coupling (k) is unity.
2. Its primary and secondary windings are pure inductors having infinitely large value.
3. Its leakage flux and leakage inductances are zero.
4. Its self- and mutual inductances are zero having no reactance or resistance.
5. Its efficiency is 100 per cent having no loss due to resistance, hysteresis, or eddy current.
6. Its transformation ratio (or turn ratio) is equal to the ratio of its secondary to primary terminal voltage and also as the ratio of its primary to secondary current.
7. Its core has permeability of infinite value.



In an ideal transformer, there is no power loss, and therefore, output must be equal to input.

That is,

$$E_2 I_2 \cos \phi = E_1 I_1 \cos \phi \quad \text{or} \quad E_2 I_2 = E_1 I_1 \quad \text{or} \quad \frac{E_2}{E_1} = \frac{I_1}{I_2}$$

Since,

$$E_2 \propto N_2; E_1 \propto N_1 \quad \text{and} \quad E_1 \cong V_1; E_2 \cong V_2$$

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2} = K (\text{transformation ratio})$$

EMF EQUATION

When sinusoidal voltage is applied to the primary winding of a transformer, a sinusoidal flux, as shown in Figure 10.12 is set up in the iron core which links with primary and secondary winding.

Let ϕ_m = Maximum value of flux in Wb

f = supply frequency in Hz (or c/s)

N_1 = No. of turns in primary

N_2 = No. of turns in secondary

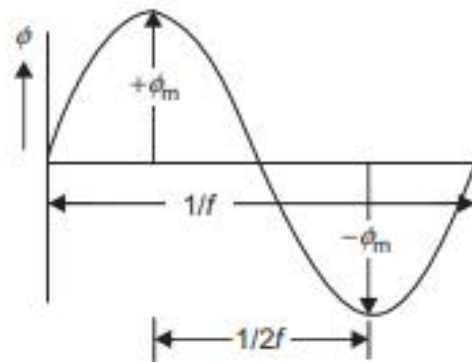


Fig. 10.12 Wave diagram of mutual flux set-up in magnetic core

As shown in Figure 10.19, flux changes from $+\phi_m$ to $-\phi_m$ in half cycle, that is, $\frac{1}{2f}$ second,

Average rate of change of flux

$$= \frac{\phi_m - (-\phi_m)}{1/2f} = 4f \phi_m \text{ Wb/s}$$

Now, the rate of change of flux per turn is the average induced emf per turn in volt.

$$\therefore \text{Average emf induced per turn} = 4f \phi_m \text{ volt}$$

For a sinusoidal wave,

$$\frac{\text{RMS value}}{\text{Average value}} = \text{Form factor} = 1.11$$

$$\therefore \text{RMS value of emf induced/turn, } E = 1.11 \times 4f \phi_m = 4.44f \phi_m \text{ volt}$$

Since primary and secondary have N_1 and N_2 turns, respectively.

$\therefore$ RMS value of emf induced in primary,

$$\begin{aligned} E_1 &= (\text{emf induced/turn}) \times \text{No. of primary turns} \\ &= 4.44N_1f \phi_m \text{ volt} \end{aligned}$$

(10.1)

Similarly, rms value of emf induced in secondary,

$$E_2 = 4.44 N_2 f \phi_m \text{ volt} \quad (10.2)$$

From eq. (10.1), we get, $\frac{E_1}{N_1} = 4.44 f \phi_m \text{ volt/turn} \quad (10.3)$

From eq. (10.1), we get, $\frac{E_2}{N_2} = 4.44 f \phi_m \text{ volt/turn} \quad (10.4)$

Equations (10.3) and (10.4) clearly show that emf induced per turn on both the sides, that is, primary and secondary is the same.

Again, we can find the voltage ratio,

$$\frac{E_2}{E_1} = \frac{4.44 N_2 f \phi_m}{4.44 N_1 f \phi_m} \quad \text{or} \quad \frac{E_2}{E_1} = \frac{N_2}{N_1} = K \text{ (transformation ratio)}$$

Equations (10.1) and (10.2) can be written in the form of maximum flux density B_m using relation,

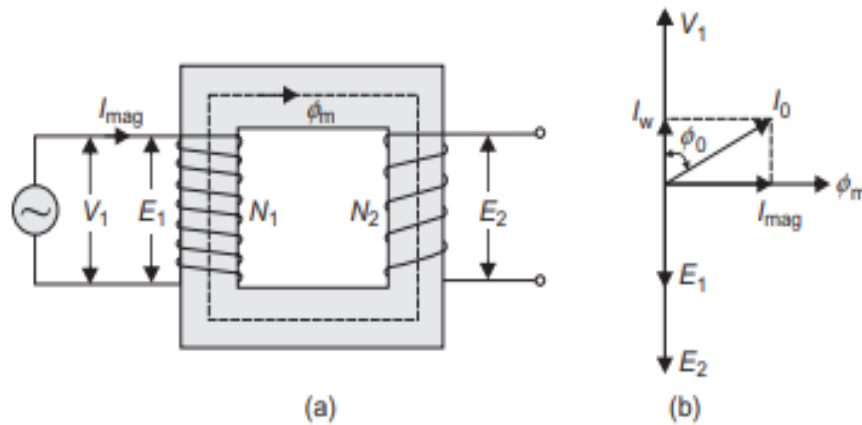
$$\phi_m = B_m \times A_i \text{ (where } A_i \text{ is iron area)}$$

$$\therefore \quad E_1 = 4.44 N_1 f B_m A_i \text{ volts} \quad \text{and} \quad E_2 = 4.44 N_2 f B_m A_i \text{ volt}$$

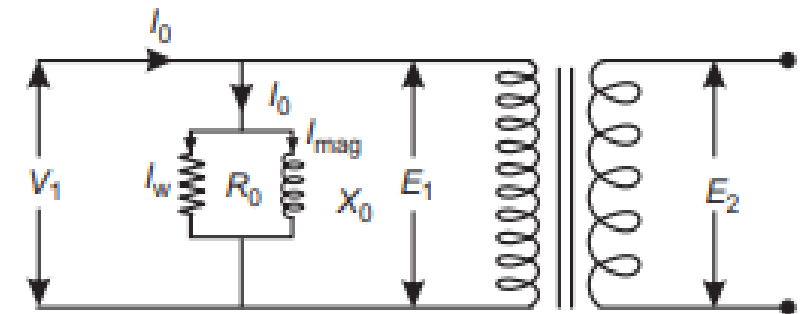
TRANSFORMER ON NO-LOAD

1. One, I_w in phase with the applied voltage V_1 , called active or working component. It supplies the iron losses and a small primary copper losses.
2. The other, I_{mag} in quadrature with the applied voltage V_1 , called reactive or magnetizing component. It produces flux in the core and does not consume any power.

From phasor (vector) diagram shown in Figure 10.13(b).

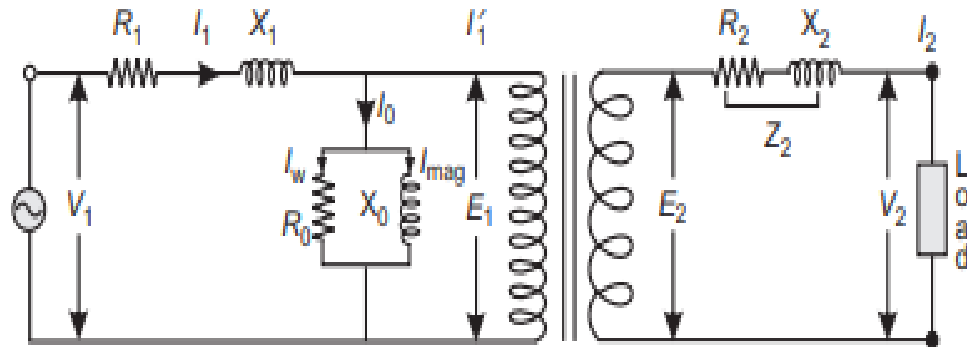
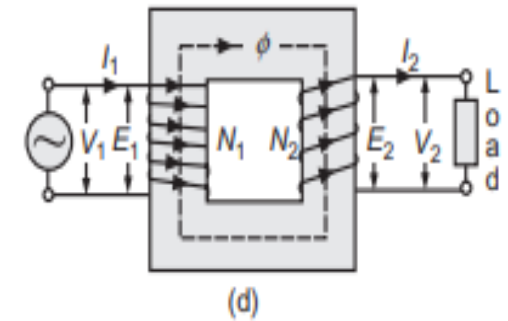
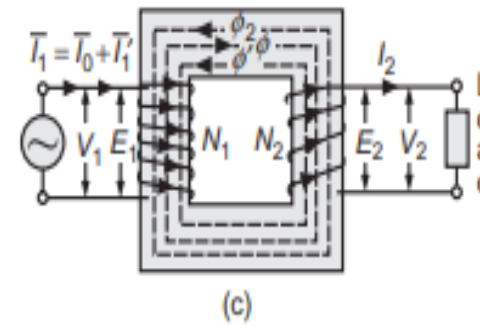
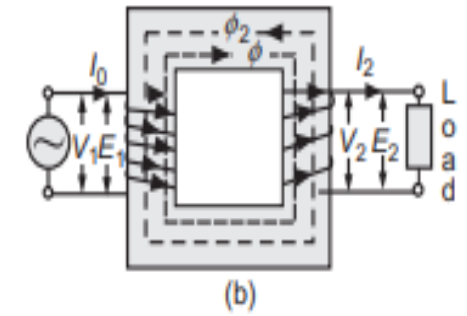
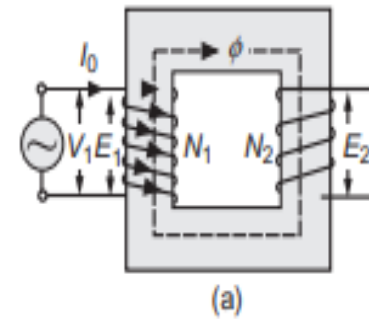
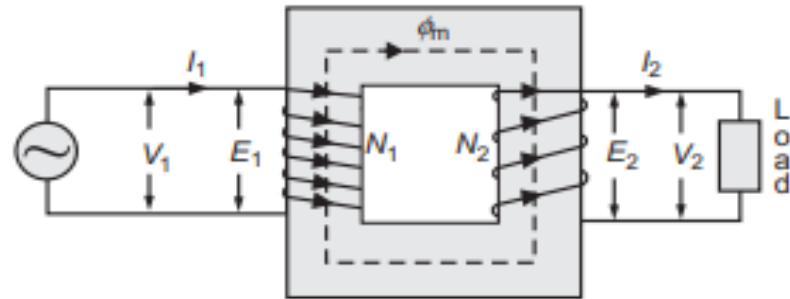


Working component,	$I_w = I_0 \cos \phi_0$
Magnetizing component,	$I_{mag} = I_0 \sin \phi_0$
No-load current,	$I_0 = \sqrt{I_w^2 + I_{mag}^2}$
Primary p.f. at no-load,	$\cos \phi_0 = \frac{I_w}{I_0}$
No-load power input,	$P_0 = V_1 I_0 \cos \phi_0$
Exciting resistance,	$R_0 = \frac{V_1}{I_w}$
Exciting reactance,	$X_0 = \frac{V_1}{I_{mag}}$



Equivalent circuit of a transformer at no-load

TRANSFORMER ON LOAD



VOLTAGE REGULATION

At a constant supply voltage, the change in secondary terminal voltage from no-load to full load with respect to no-load voltage is called voltage regulation of the transformer.

Let, E_2 = Secondary terminal voltage at no-load.

V_2 = Secondary terminal voltage at full load.

Then, voltage regulation $= \frac{E_2 - V_2}{E_2}$ (per unit)

In the form of percentage, $\% \text{ Reg} = \frac{E_2 - V_2}{E_2} \times 100$

When all the quantities are referred to the primary side of the transformer

$$\% \text{ Reg} = \frac{V_1 - E_1}{V_1} \times 100$$

LOSSES IN A TRANSFORMER

1. **Core or iron losses:** When AC supply is given to the primary winding of a transformer, an alternating flux is set up in the core; therefore, hysteresis and eddy current losses occur in the magnetic core.

(a) **Hysteresis loss:** When the magnetic material is subjected to reversal of flux, power is required for the continuous reversal of molecular magnets. This power is dissipated in the form of heat and is known as hysteresis loss ($P_h = K_h V f B_m^{1.6}$). This loss can be minimized by using silicon steel material for the construction of core.

(b) **Eddy current loss:** Since flux in the core of a transformer is alternating, it links with the magnetic material of the core itself also. This induces an emf in the core and circulates eddy currents. Power is required to maintain these eddy currents. This power is dissipated in the form of heat and is known as eddy current loss ($P_e = K_e V f^2 t^2 B_m^2$). This loss can be minimized by making the core of thin laminations.

It is already seen in article-8 that the flux set up in the core of the transformer remains constant from no-load to full load. Hence, iron loss are independent of the load and are known as constant losses.

2. **Copper losses:** Copper losses occur in both primary and secondary windings due to their ohmic resistance. If I_1, I_2 are the primary and secondary currents and R_1, R_2 are the primary and secondary resistances, respectively.

$$\text{Then, total copper losses} = I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{ep} = I_2^2 R_{es}$$

EFFICIENCY OF A TRANSFORMER

The efficiency of a transformer is defined as the ratio of output to the input power, the two being measured in same units (either in watts or in kW).

$$\text{Transformer efficiency, } \eta = \frac{\text{Output power}}{\text{Input power}} = \frac{\text{Output power}}{\text{Output power} + \text{Losses}}$$

$$\text{or } \eta = \frac{\text{Output power}}{\text{Output power} + \text{Iron losses} + \text{Copper losses}} = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + P_c}$$

Where V_2 = Secondary terminal voltage

I_2 = Full-load secondary current

$\cos \phi_2$ = p.f. of the load

P_i = Iron losses = Hysteresis losses + eddy current losses (constant losses)

P_c = Full-load copper losses = $I_2^2 R_{es}$ (variable losses)

If x is the fraction of the full load, the efficiency of the transformer at this fraction is given by the relation

$$\eta_x = \frac{x \times \text{Output at full load}}{x \times \text{Output at full load} \times P_i + x^2 P_c} = \frac{x V_2 I_2 \cos \phi_2}{x V_2 I_2 \cos \phi_2 + P_i + x^2 I_2^2 R_{es}}$$

The copper losses vary as the square of the fraction of the load.

CONDITION FOR MAXIMUM EFFICIENCY

The efficiency of a transformer at a given load and p.f. is expressed by the relation

$$\eta = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{es}} = \frac{V_2 \cos \phi_2}{V_2 \cos \phi_2 + P_i / I_2 + I_2 R_{es}}$$

$$\frac{d}{dI_2} \left(V_2 \cos \phi_2 + \frac{P_i}{I_2} + I_2 R_{es} \right) = 0$$

or

$$0 - \frac{P_i}{I_2^2} + R_{es} = 0 \quad \text{or} \quad I_2^2 R_{es} = P_i$$

that is,

$$\text{Copper losses} = \text{iron losses}$$

Thus, the efficiency of a transformer will be maximum when copper (or variable) losses are equal to iron (or constant) losses.

$$\therefore \eta_{\max} = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + 2P_i} \quad (\text{since } P_c = P_i)$$

From equation (ii), the value of output current I_2 at which the efficiency of the transformer will be maximum is given by

$$I_2 = \sqrt{\frac{P_i}{R_{es}}}$$

If x is the fraction of full-load kVA at which the efficiency of the transformer is maximum.

Then, copper losses = $x^2 P_c$ (where P_c is the full-load Cu losses)

Iron losses = P_i

$$\text{For maximum efficiency,} \quad x^2 P_c = P_i; \quad x = \sqrt{\frac{P_i}{P_c}}$$

$\therefore$ Output kVA corresponding to maximum efficiency

$$= x \times \text{full-load kVA} = \text{full-load kVA} \times \sqrt{\frac{P_i}{P_c}}$$

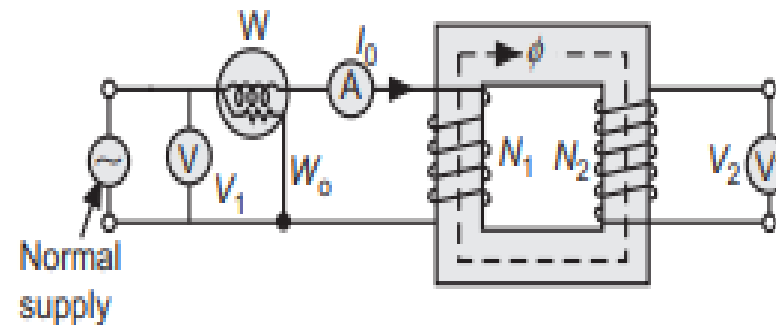
$$= \text{full-load kVA} \times \sqrt{\frac{\text{Iron losses}}{\text{Copper losses at full load}}}$$

ALL-DAY EFFICIENCY

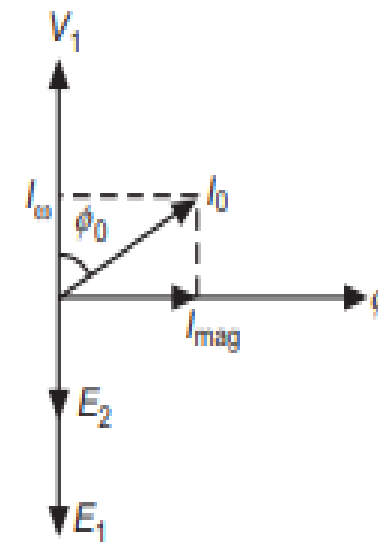
$$\therefore \text{All day efficiency} = \eta_{\text{all day}} = \frac{\text{Output in kWh}}{\text{Input in kWh}} \quad (\text{for 24 h})$$

To find this all-day efficiency, we have to know the load cycle on the transformer.

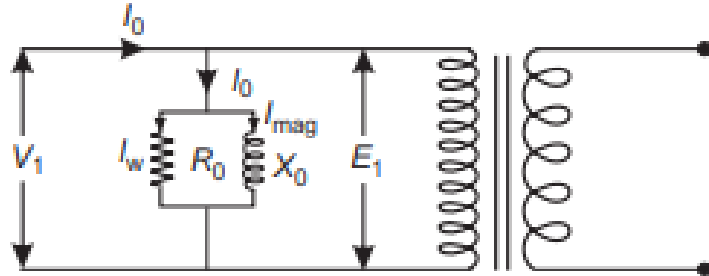
Open-circuit or No-load Test



(a)



(b)



The ratio of voltmeter readings, $\frac{V_2}{V_1}$ gives the transformation ratio of the transformer. The phasor diagram of transformer at no-load is shown in Figure 10.31(b).

Let the wattmeter reading = W_0

voltmeter reading = V_1

and ammeter reading = I_0

Then, iron losses of the transformer $P_i = W_0$

that is, $V_1 I_0 \cos \phi_0 = W_0$

No-load power factor,

$$\cos \phi_0 = \frac{W_0}{V_1 I_0}$$

Working component,

$$I_w = \frac{W_0}{V_1} \quad (I_w = I_0 \cos \phi_0)$$

Magnetising component

$$I_{\text{mag}} = \sqrt{I_0^2 - I_w^2}$$

No-load, parameters, that is,

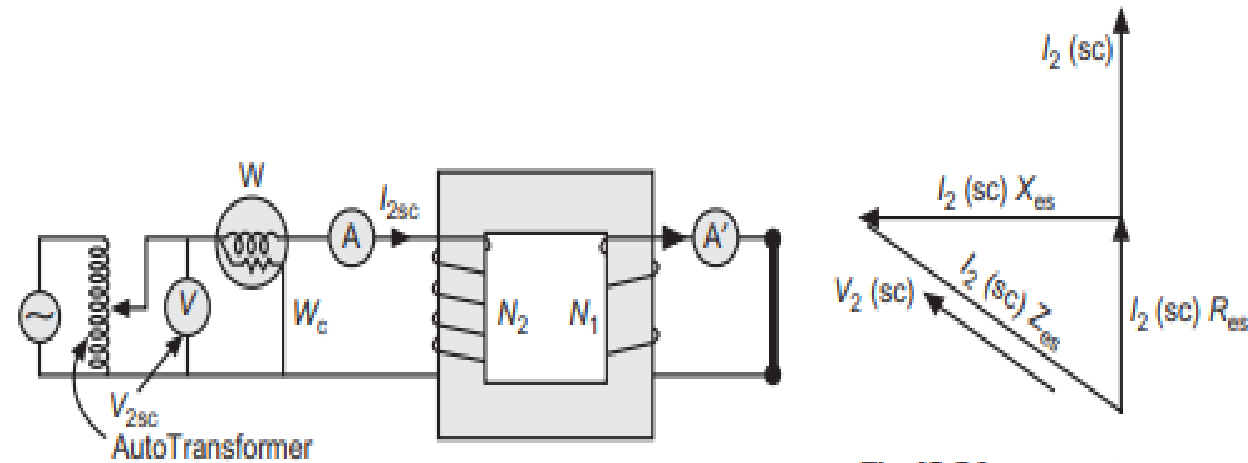
$$\text{Equivalent exciting resistance, } R_0 = \frac{V_1}{I_w}$$

$$\text{Equivalent exciting reactance, } X_0 = \frac{V_1}{I_{\text{mag}}}$$

Short Circuit Test

This test is carried out to determine the following:

1. Copper losses at full load (or at any desired load). These losses are required for the calculations of efficiency of the transformer.
2. Equivalent impedance (Z_{es} or Z_{ep}), resistance (R_{es} or R_{ep}) and leakage reactance (X_{es} or X_{ep}) of the transformer referred to the winding in which the measuring instruments are connected. Knowing equivalent resistance and reactance, the voltage drop in the transformer can be calculated and hence regulation of transformer is determined.



This test is usually carried out on the high-voltage side of the transformer, that is, a wattmeter W, voltmeter V, and an ammeter A are connected in high-voltage* winding (say secondary). The other winding (primary) is then short circuited by a thick strip or by connecting an ammeter A' across the terminals

The iron losses are negligibly small due to low value of flux as these losses are approximately proportional to the square of the flux. Hence, wattmeter reading W_c only represents the copper losses in the transformer windings for all practical purposes. The applied voltage V_{2sc} is measured by the voltmeter V which circulates the current I_{2sc} (usually full-load current) in the impedance Z_{cs} of the transformer to the side in which instruments are connected

Let the wattmeter reading = W_c

voltmeter reading = V_{2sc}

and ammeter reading = I_{2sc}

Then, full-load copper losses of the transformer,

$$P_c = \left(\frac{I_{2fl}}{I_{2sc}} \right)^2 W_c \quad \text{and} \quad I_{2sc}^2 R_{cs} = W_c$$

Equivalent resistance referred to secondary, $R_{cs} = \frac{W_c}{I_{2sc}^2}$

From phasor diagram as shown in Figure 10.34

$$I_{2sc} Z_{cs} = V_{2sc}$$

∴ Equivalent impedance referred to secondary, $Z_{cs} = V_{2sc} / I_{2sc}$

Equivalent reactance referred to secondary, $X_{cs} = \sqrt{(Z_{cs})^2 - (R_{cs})^2}$



THANKS